ROOTS : AN AI -POWERED MEMORY COMPANION FOR THE COLLEGE TRANSITION

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Abstract

The transition to college can involve separation from familiar social environments, changes in daily routines, and the challenge of forming new relationships. Digital tools can support individual aspects of this transition, including memory storage, mood tracking, or social interaction, but these functions are often addressed separately. This paper presents ROOTS, a mobile-first memory companion designed to bring personal memory management, mood reflection, shared memory spaces, and connection-oriented activities into a single application. The system follows a modular client–server architecture comprising a React Native frontend developed with Expo and TypeScript, a Python backend implemented using FastAPI, and a SQLite database accessed through SQLAlchemy. Its backend consists of five modules: memory management, mood tracking, connection games, memory capsules, and an AI engine. The implemented AI engine applies keyword-based sentiment and emotion analysis to text and uses rule-based logic to generate memory-reel titles, emotional tones, and template-based narration from recent memories. The system also provides a five-level mood check-in mechanism and invite-code-based memory capsules. Functional validation was conducted through six automated API tests using pytest, supplemented by manual endpoint checks and application demonstrations. The tests cover core workflows involving memory operations, mood tracking, game sessions, capsules, and AI endpoints. The current implementation remains a prototype: voice transcription is represented by a placeholder, image captioning and large language

model-based narration are not implemented, and the application has not undergone a user study. The paper describes the architecture, implemented processing methods, functional validation, and limitations of an integrated memory-centred application for the college transition.

Keywords: Memory Companion · College Transition · Sentiment Analysis · Emotion Analysis · Mood Tracking · Mobile Application

1 Introduction

1.1 Background and Motivation

The transition from school to college represents a significant change in students' social environments, daily routines, and personal responsibilities. Students may need to adapt to unfamiliar surroundings, establish new relationships, and maintain connections with family and friends who are no longer physically nearby. For some students, these changes can be accompanied by homesickness, loneliness, and difficulties adjusting to university life. Previous research has examined homesickness as a distinct experience associated with separation from familiar environments and the process of adapting to a new setting [1]. Research on first-year international students has also explored the relationship between acculturative stress and adjustment during the initial period of university life [7].

Digital applications offer several ways to preserve personal experiences, communicate with others, and reflect on everyday activities. However, memory recording, mood tracking, and social engagement are often treated as separate functions. Personal photographs and written memories may preserve experiences, while mood-tracking applications record emotional states and interactive applications provide opportunities for engagement. Bringing these functions together presents an opportunity to create a more integrated digital experience for students transitioning to college.

ROOTS, an AI-powered memory companion, is designed around this opportunity. It provides a unified application for recording personal memories, tracking moods, participating in connection-oriented games, and organizing shared experiences through memory capsules. An AI processing component further supports memory reflection by analyzing textual memories and generating rule-based memory reels. The system aims to provide a convenient digital space in which students can preserve meaningful experiences while engaging with both personal and shared memories.

1.2 Problem Statement

Existing digital tools address different aspects of personal memory preservation, emotional reflection, and social interaction. However, managing these activities across separate platforms may fragment the user's experience and require information to be maintained in multiple places. A student might record an experience in one application, track a mood in another, and use a separate platform to share memories with friends.

The problem addressed in this work is the design and implementation of an integrated mobile application that combines memory management, structured mood tracking, shared memory collections, connection-oriented activities, and lightweight AI-assisted memory reflection. In addition to providing these functions, the application must coordinate data storage, backend processing, and user interactions within a modular architecture.

The scope of the present work is the implementation and functional validation of such a system. It does not assume that integrating these features automatically reduces homesickness, improves mental health, or strengthens friendships; these outcomes require separate empirical evaluation.

1.3 Research Gap

Prior research has examined several relevant areas, including homesickness and university adjustment [1, 7], the relationship between social media use and loneliness [2], AI-based conversational support for college students [3], AI-assisted reminiscence [4], and interventions involving student friendships and social relationships [5, 6]. These studies provide useful foundations for understanding student adjustment, digital support, memory-based interaction, and social connection.

Nevertheless, the technical implementation challenge of combining personal memory management, structured mood check-ins, shared memory collections, interactive activities, and automated memory-reel generation within one application remains relevant. A unified system must coordinate these features while ensuring that its AI-generated outputs are grounded in the available memory content and that its limitations are clearly understood.

ROOTS addresses this implementation-oriented gap by bringing these functions into a single mobile-first application with a modular backend and a lightweight, rule-based AI component. The contribution is the integration and implementation of these features rather than a claim that no comparable application exists or that the system has demonstrated superior psychological outcomes.

1.4 Research Objectives

The primary objective of this work is to design, implement, and functionally validate ROOTS as an integrated memory companion for the college transition. The specific objectives are:

1. To develop a mobile application that supports the creation, retrieval, and organization of personal memories.
2. To implement structured mood check-ins and mood-history retrieval.
3. To provide connection-oriented activities and invite-code-based shared memory capsules.
4. To implement lightweight sentiment analysis, keyword-based emotion identification, and rule-based memory-reel generation.
5. To integrate these capabilities through a modular backend and persistent database.
6. To validate the core application workflows through automated API tests and manual endpoint checks while documenting the limitations of the current prototype.

1.5 Contributions

The contributions of this work are as follows:

- Integrated application design: ROOTS combines personal memory management, mood tracking, connection games, and shared memory capsules within a single mobile application.
- Modular system implementation: The system uses a React Native and Expo frontend, a FastAPI backend, and SQLite persistence through SQLAlchemy.
- Lightweight AI processing: A rule-based pipeline performs sentiment scoring, identifies emotional categories using predefined keyword dictionaries, and generates memory reels through predefined rules and narrative templates.
- Functional validation: Automated tests and manual endpoint checks are used to examine selected application workflows, including memory operations, mood tracking, game sessions, capsules, and AI endpoints.
- Explicit implementation boundaries: The work documents the current limitations of the prototype, including the absence of a user study, quantitative AI accuracy evaluation, integrated image captioning, and LLM-based narration.

Together, these contributions establish a technical foundation for further investigation into integrated, memory-centred digital applications for students experiencing the college transition.

2 Related Work

2.1 Homesickness and Adjustment During the College Transition

Homesickness and adjustment to unfamiliar environments are important considerations when examining the transition to university life. Stroebe et al. [1] presented a systematic review of the scientific literature on homesickness, examining its conceptualization and its relationship with adaptation to a new environment. Their work provides a foundation for understanding the challenges that can arise when individuals move away from familiar surroundings.

Similarly, Koo et al. [7] investigated acculturative stress and adjustment among first-year international college students through a longitudinal study. Their work highlights the relevance of examining students' experiences over the course of their initial adjustment period.

These studies motivate the need for digital tools that help students preserve meaningful personal experiences and reflect on their changing circumstances. ROOTS approaches this need through memory recording, mood-history tracking, and memory-reel generation. However, the current implementation is intended to support memory reflection; it has not been evaluated as an intervention for homesickness or adjustment difficulties.

2.2 Social Media, Loneliness, and Digital Emotional Support

Digital communication platforms provide opportunities for maintaining relationships and interacting with others. However, the relationship between digital engagement and emotional well-being is complex. Hill and Merianos [2] explored excessive social media use in relation to loneliness among college students in the United States. Their work provides relevant context for considering how digital interaction relates to students' social experiences.

Nyakhar and Wang [3] examined the effectiveness of artificial intelligence chatbots for mental health and well-being among college students through a rapid systematic review. Their work represents a related direction in the use of AI-enabled systems for student support.

ROOTS differs in its immediate technical focus. Rather than functioning primarily as a conversational chatbot or a social media platform, it combines personal memory storage, structured mood check-ins, and shared memory activities with lightweight text analysis. Its AI component analyzes recorded memories and produces template-based memory reels instead of providing conversational counselling or clinical recommendations. This distinction defines the scope of the current system and avoids positioning it as a replacement for professional support.

2.3 AI-Assisted Reminiscence and Personal Memory Systems

AI-assisted reminiscence is a relevant area of research for applications that organize personal experiences and facilitate reflection. Wang et al. [4] investigated the development and usability of an AI-driven interactive multimodal photo album designed to promote personalized reminiscence among cognitively intact older adults. Their work illustrates how AI-based systems can support engagement with personal memories through an interactive digital format.

ROOTS similarly treats recorded experiences as inputs for personalized memory presentation. However, its current implementation operates primarily on textual memory content. It applies keyword-based sentiment analysis and emotion identification, then uses predefined rules to select memory-reel titles and emotional tones. The resulting narrative is generated from templates rather than a large language model.

The distinction is important because the current ROOTS prototype does not yet implement automated image captioning or multimodal memory interpretation. The work therefore focuses on the integration of textual memory analysis with a broader application architecture, while multimodal memory processing remains a direction for future development.

2.4 Social Connection and Friendship Formation

Social relationships play an important role in students' university experiences. Boda et al. [5] investigated the short-term and long-term effects of a social-network intervention on friendships among university students. Erb et al. [6] reviewed the importance of college roommate relationships and developed a conceptualization of factors relevant to these relationships.

These studies provide context for considering how digital systems might facilitate engagement with shared experiences and relationships. ROOTS incorporates connection-oriented games and memory capsules to support interactive engagement and the collection of shared memories. The games provide activities centred on memory matching and chronological ordering, while capsules allow memories to be grouped into shared collections through invite codes.

Nevertheless, these features should be distinguished from a validated social-network intervention. The present work implements mechanisms for shared memory interaction but does not establish that using them improves friendship formation, relationship quality, or social integration. Evaluating such outcomes would require a dedicated user study.

Table 1 Comparative Analysis of Related Studies and ROOTS

Study	**Research Focus**	**Relevance to ROOTS**	**Limitation Relative to the Present Work**
Stroebe et al. [1]	Homesickness during life transitions	Establishes the emotional-adjustment context	Does not evaluate ROOTS or its architecture
Hill and Merianos [2]	Social media use and loneliness among college students	Motivates consideration of digital social experiences	Association does not establish causality or the effectiveness of ROOTS
Nyakhar and Wang [3]	AI chatbots for student mental health and well-being	Provides context for AI-supported digital tools	Focuses on chatbot-based support rather than ROOTS's memory-centred design
Wang et al. [4]	AI-driven interactive photo albums for reminiscence	Informs the motivation for structured memory revisiting	ROOTS currently uses text analysis and template-based narration, not multimodal photo understanding
Boda et al. [5]	Social network intervention and friendship formation	Motivates investigation of structured social interaction	Does not validate ROOTS's games or capsules
Erb et al. [6]	College roommate relationships	Supports the relevance of shared experiences among roommates	Does not evaluate the ROOTS capsule mechanism
Koo et al. [7]	Acculturative stress and adjustment among international students	Provides context for first-year transition challenges	Does not evaluate a digital memory companion

2.5 Research Gap and Positioning of ROOTS

The reviewed literature covers several related research directions: homesickness and student adjustment [1, 7], social media use and loneliness [2], AI-based support for college students [3], AI-assisted reminiscence [4], and friendship formation and roommate relationships [5, 6]. Together, these works establish the relevance of memory, emotional reflection, digital interaction, and social relationships to the broader context of student experiences.

ROOTS builds on these directions through an implementation-oriented contribution: the integration of personal memory management, structured mood tracking,

connection games, shared memory capsules, and rule-based AI memory-reel generation within a single mobile-first application.

The current system is distinguished by its combination of these modules and its lightweight technical implementation rather than by a claim of superior AI performance or demonstrated psychological effectiveness. Its functional validation examines selected backend and API workflows, while the accuracy of its sentiment and emotion analysis, usability among college students, and effects on social or emotional outcomes remain to be evaluated.

3 Proposed System and Architecture

3.1 System Overview

ROOTS is an AI-powered memory companion designed to support students during the transition to college by integrating personal memory management, mood tracking, social connection activities, and shared memory spaces within a single mobile application. The system provides a centralized platform for recording personal experiences, revisiting meaningful moments, reflecting on emotional states, and organizing shared memories.

The application follows a modular client–server architecture. Its frontend is developed using React Native, Expo, and TypeScript, while the backend is implemented using Python and FastAPI. SQLite is used for persistent data storage, with SQLAlchemy providing the object-relational mapping layer. The backend exposes REST API endpoints through which the frontend accesses the application's core functionality.

The system consists of five principal modules: memory management, mood tracking, connection games, memory capsules, and AI processing. Each module performs a specific set of operations while interacting with other components where necessary. This modular design separates user-interface operations, application logic, AI processing, and data persistence.

The AI component currently performs keyword-based sentiment analysis, dictionary-based emotion identification, and rule-based memory-reel generation. It processes textual memories to estimate sentiment polarity, identify relevant emotional categories, and generate titles, tones, and template-based narratives. The current implementation does not rely on a large language model for narrative generation.

3.2 System Architecture

The ROOTS architecture comprises five logical layers.

User Interaction Layer: This layer represents the student's interaction with the application. Users can create and revisit memories, record mood check-ins, participate in connection games, and create or join shared memory capsules.

Frontend Layer: The frontend uses React Native, Expo, and TypeScript to provide the application's screens and user interactions. It sends API requests to the backend for operations that require server-side processing or persistent storage and presents the responses to the user.

Backend Service Layer: The backend is implemented using FastAPI and exposes REST endpoints for the application's core modules. It validates incoming requests, executes application logic, and coordinates data access through SQLAlchemy.

AI Processing Layer: This layer analyzes available memory text using predefined keyword dictionaries, simple negation handling, and rule-based conditions. It produces sentiment scores, identifies emotional categories, and generates memory reels using predefined titles, tones, and narrative templates.

Data Storage Layer: SQLite provides persistent storage for the application's supported records. SQLAlchemy facilitates database operations for memories, mood entries, game sessions, and memory capsules.

The separation of these layers enables individual components to be maintained and extended independently. For example, a more advanced sentiment analysis model could eventually replace the current dictionary-based implementation without requiring the entire application architecture to be redesigned.

3.3 Core System Modules

3.3.1 Memory Management Module

The memory management module supports the creation, retrieval, listing, and deletion of personal memory entries. It validates supported operations and allows memories to be optionally associated with a memory capsule.

Memory text also serves as an input to the AI processing component. When memory analysis or memory-reel generation is requested, the available text can be processed to estimate sentiment polarity and identify emotional categories.

The module therefore supports both memory storage and AI-assisted reflection. The current implementation focuses on textual memories; automated image captioning and extraction of descriptions from photographs have not yet been implemented.

3.3.2 Mood Tracking Module

The mood tracking module allows users to record their emotional state through five predefined categories: anxious, low, neutral, content, and energized. These categories are associated with application-specific numerical scores of 2, 4, 6, 8, and 10, respectively.

The module supports mood recording, retrieval of mood history, and summary information. These functions allow users to revisit previously recorded check-ins.

The numerical mapping is an internal application design and is not a clinically validated psychological scale. Furthermore, mood check-ins and memory-based sentiment analysis are not yet fully integrated, so the current system does not establish a direct relationship between recorded moods and the emotional content of individual memories.

3.3.3 Connection Games Module

The connection games module provides interactive activities centred on memories and shared experiences. The application includes Memory Match and Timeline Challenge.

Memory Match provides a memory-oriented matching activity, while Timeline Challenge focuses on arranging events chronologically. These activities complement the application's memory-management functionality by offering additional ways to engage with experiences.

The backend supports persistence of Memory Match game sessions. Timeline Challenge currently runs locally and does not persist its results through the backend. The two activities therefore differ in their current data-persistence capabilities.

Although these games are intended to encourage engagement, their effects on social connection, loneliness, or student adjustment have not been evaluated through a user study.

3.3.4 Memory Capsules Module

The memory capsules module allows memories to be organized into shared collections. Users can create capsules and join existing ones using unique invite codes, providing a mechanism for collecting memories associated with shared experiences.

This module extends the application beyond individual memory storage by supporting shared memory spaces. However, the current implementation does not include a complete authentication system, detailed membership management, or fine-grained permissions for individual memories. Capsule sharing should therefore be understood as a prototype capability rather than a production-ready secure-sharing mechanism.

Table 2 ROOTS System Modules and Their Responsibilities

Module	**Main Responsibility**	**Technology / Approach**
Memory Management	Create, retrieve, and delete memories	FastAPI, SQLite
Mood Tracking	Record mood check-ins	Predefined mood categories
Connection Games	Support interactive memory games	Frontend logic and backend sessions
Memory Capsules	Group memories using invite codes	REST APIs and database
AI Engine	Analyse text and generate memory reels	Keyword-based rules and templates

3.3.5 AI Processing Module

The AI processing module provides three principal functions: sentiment analysis, emotion identification, and memory-reel generation.

Sentiment analysis uses predefined positive and negative keyword dictionaries, simple negation handling, and selected word-prefix matching to estimate the polarity of textual memories. The resulting score is constrained to the interval from -1 to +1.

Emotion identification uses predefined keyword dictionaries to identify categories such as joy, nostalgia, calm, gratitude, pride, reflection, anxiety, fatigue, and hope.

Memory-reel generation processes recent memories, calculates their average sentiment polarity, and applies predefined rules to select an appropriate title and emotional tone. The default configuration processes up to seven recent memories, with a configurable maximum of fifty. The generated reel includes template-based narration and highlights recent memories.

The module provides a lightweight method of generating memory presentations without requiring external generative-model inference. However, its outputs depend on predefined vocabulary and rules, which may not adequately capture nuanced language, sarcasm, unfamiliar expressions, or emotions that are not represented in its dictionaries.

3.4 Data Flow and Module Interaction

The system follows a request–response workflow for operations involving the backend.

1. User input: The user performs an operation through the frontend, such as creating a memory, recording a mood, joining a capsule, or requesting a memory reel.
2. API request: For backend-supported operations, the frontend sends a request to the appropriate FastAPI endpoint.
3. Request processing: The backend validates the request and routes it to the relevant functional module.
4. AI processing: Where required, the AI engine analyzes available memory text using its keyword-based sentiment and emotion-processing methods and applies the rules used for memory-reel generation.
5. Database interaction: The backend uses SQLAlchemy to store or retrieve records from SQLite when persistence is required.
6. Response generation: The backend returns the operation's result through the API response.
7. User presentation: The frontend displays the returned information to the user.

Not every operation requires all stages. Some game interactions are handled locally, whereas memory operations and other persistent features require backend communication. Similarly, the current AI component uses predefined rules and templates rather than an external large language model.

3.5 Design Considerations

3.5.1 Modularity and Maintainability

The separation of functionality into five backend modules supports maintainability and future development. Individual components can be modified or extended without requiring all other modules to be rewritten.

3.5.2 Lightweight Processing

The current implementation uses FastAPI, SQLite, and rule-based AI processing. Its implemented text-analysis pipeline does not require a dedicated GPU or external generative-model service, making it suitable for a lightweight prototype.

3.5.3 Usability and Interpretability

The application combines memory storage, mood check-ins, shared collections, and interactive activities in one interface. The rule-based AI pipeline also makes it relatively straightforward to inspect how keyword detection and polarity thresholds influence memory-reel selection. Whether these design choices improve usability requires further evaluation.

3.5.4 Privacy and Access Control

Personal memories and mood entries may contain sensitive information. The current prototype does not implement a complete authentication and authorization framework or fine-grained access control. Secure data isolation, membership permissions, and appropriate deployment safeguards are necessary before using the application with real users at scale.

3.5.5 Extensibility

The architecture provides a foundation for future capabilities, including speech transcription, image captioning, language-model-based narration grounded in memory content, and stronger integration between mood history and memory analysis. These remain future extensions rather than completed features.

4 AI Methodology

4.1 Overview of the AI Pipeline

The AI component of ROOTS is designed to transform textual memories into structured sentiment information, emotion categories, and personalized memory-reel presentations. The current implementation uses a lightweight, rule-based approach rather than a trained machine-learning classifier or a large language model.

The pipeline consists of three principal stages: sentiment analysis, emotion identification, and memory-reel generation. The sentiment-analysis component estimates the polarity of memory text using predefined keyword dictionaries and simple negation handling. The emotion-identification component searches for predefined emotional keywords. The memory-reel generator then uses information from recent memories to select a suitable title and emotional tone and produce a template-based narrative.

The overall processing pipeline is illustrated conceptually as follows:

Textual Memories → Sentiment Analysis and Emotion Identification → Aggregation of Recent Memory Scores → Rule-Based Reel Selection → Template-Based Narrative Generation

This design prioritizes simplicity and interpretability. However, the output quality depends on the coverage of the predefined dictionaries and the effectiveness of the rules used to interpret the text.

4.2 Sentiment Analysis

4.2.1 Keyword-Based Sentiment Identification

The sentiment-analysis component estimates the emotional polarity of a memory by identifying predefined positive and negative keywords. The implementation uses keyword dictionaries rather than a model trained on a labelled dataset.

Positive terms contribute positively to the sentiment score, whereas negative terms contribute negatively. The implemented scoring procedure applies a contribution of 0.2 for recognized positive or negative terms, with simple negation handling used to adjust the interpretation of recognized keywords. Selected word prefixes are also checked to recognize certain variations of positive and negative expressions.

For example, a memory containing expressions associated with happiness or excitement may receive a positive polarity score, whereas text containing expressions associated with stress or anxiety may receive a negative score. These examples illustrate the intended behavior of the dictionary-based method and do not represent measured model performance.

4.2.2 Negation Handling

A simple negation-handling mechanism is incorporated to account for cases in which a recognized sentiment keyword is preceded by a negation term. The implementation checks a limited preceding context, covering one or two preceding words, against a predefined set of negation terms.

This mechanism attempts to distinguish between expressions such as a positive statement and a negated positive statement. However, it does not provide comprehensive linguistic analysis. Complex sentence structures, long-distance dependencies, sarcasm, and context-dependent meanings may not be interpreted correctly.

4.2.3 Sentiment Score Calculation

The contributions from recognized keywords are combined to obtain a polarity score. The resulting value is rounded to two decimal places and constrained to the interval $[-1, 1]$.

The score can be interpreted as follows: - Values closer to $+1$ indicate a stronger positive polarity under the implemented rules. - Values closer to -1 indicate a stronger negative polarity under the implemented rules. - Values closer to zero indicate a comparatively neutral or mixed result under the implemented scoring procedure. These interpretations describe the behavior of the application's scoring mechanism. The score is not a calibrated probability, a validated measure of emotional intensity, or a clinical assessment.

The implementation provides a computationally inexpensive method for assigning polarity to textual memories. However, because the method relies on predefined keywords and fixed contributions, its reliability across different writing styles and contexts has not yet been quantitatively evaluated.

4.3 Emotion Classification

In addition to sentiment polarity, ROOTS identifies emotional categories using predefined keyword dictionaries. Whereas sentiment analysis estimates polarity along a positive-to-negative range, emotion identification associates memory text with specific emotional categories.

The implemented categories include: - Joy: Expressions associated with happiness and positive experiences. - Nostalgia: Expressions associated with recalling familiar people, places, or past experiences. - Calm: Expressions associated with peaceful or relaxed experiences. - Gratitude: Expressions associated with appreciation. - Pride: Expressions associated with accomplishment or personal achievement. - Reflection: Expressions associated with thinking about experiences or personal growth. - Anxiety: Expressions associated with worry or apprehension. - Fatigue: Expressions associated with tiredness or exhaustion. - Hope: Expressions associated with optimism or future expectations. The component identifies categories by matching the available text against the corresponding predefined vocabulary. The resulting categories provide additional information that can be used by the memory-reel generation component.

Sentiment polarity and emotion categories serve complementary purposes. A polarity score represents the overall direction estimated by the scoring rules, whereas an emotion category provides a more specific label when corresponding keywords are recognized. These outputs should not be treated as interchangeable: a negative polarity score does not, by itself, determine whether a memory reflects anxiety, fatigue, or another emotional state.

The current method is limited by its dictionary-based design. It may fail to recognize an emotion expressed using unfamiliar vocabulary, implicit meaning, or language that does not contain the predefined keywords. Furthermore, the implementation has not been evaluated against manually annotated emotional labels, so classification accuracy and category-level performance remain unknown.

4.4 Memory-Reel Generation

4.4.1 Selection and Aggregation of Memories

The memory-reel generation component creates a consolidated presentation from recent memory entries. It processes seven recent memories by default and supports a configurable maximum limit of fifty memories.

For the selected memories, the component uses the available sentiment polarity scores to calculate an average polarity. If the individual scores are represented by

$s_1, s_2, \ldots, s_n$, the average sentiment polarity is calculated as:

$$\bar{s} = \frac{1}{n} \sum_{i=1}^{n} s_i$$

where n represents the number of memories included in the aggregation and s_i represents the polarity score associated with the i-th memory.

The average provides a summary of the polarity scores under the implemented rules. It does not represent a validated measurement of the user's overall emotional state. In particular, averaging scores may conceal differences between individual memories, and the result depends on the memories selected for the reel.

4.4.2 Rule-Based Tone and Title Selection

After calculating the average polarity, the generator applies predefined conditions to select the memory reel's title and emotional tone. The current implementation uses the following decision rules.

Table 3 Rule-Based Selection of Memory-Reel Tone and Title

Condition	**Selected Tone**	**Reel Title**
Average polarity ≥ 0.3	Uplifting & Connected	Golden Moments & Growth
Average polarity ≤ -0.2	Gentle Resilience	Navigating Challenges & Moving Forward
Neither threshold is met, and nostalgia is identified	Warm & Nostalgic	Familiar Echoes of the Past
Neither threshold is met, and calm is identified	Peaceful & Grounded	Quiet Reflections
No preceding condition applies	Thoughtful Balance	Fragments of a Beautiful Week

The conditions provide a deterministic mapping between the calculated polarity, recognized emotional categories, and the resulting reel presentation. The nostalgia and calm conditions provide additional selection rules when neither polarity threshold is met. If none of the specified conditions applies, the generator uses the default tone and title.

This mechanism allows the system to produce a consistent presentation without requiring generative-model inference. However, the selected title and tone reflect the application's predefined decision rules rather than a comprehensive understanding of the user's experiences.

4.4.3 Template-Based Narrative Generation

After selecting the tone and title, the system generates a narrative using predefined templates. The narrative is constructed from the available memory information and

the selected presentation category. The generator also highlights three recent memories as part of the resulting reel.

Unlike a large language model, the template-based generator does not compose unrestricted natural-language narratives through learned language generation. Its output is constrained by the templates and rules implemented in the application.

This approach reduces the need for external inference services and makes the generation process easier to inspect. However, it limits narrative variety and may produce generic descriptions when the selected memories contain complex or diverse experiences. The current implementation should therefore be understood as a rule-based memory-presentation mechanism rather than a fully generative conversational system.

4.5 Integration of AI Components

The sentiment-analysis, emotion-identification, and memory-reel-generation components form a sequential processing pipeline.

First, the system obtains the relevant textual memories. The sentiment-analysis component calculates polarity scores using the predefined keyword dictionaries, contribution rules, and negation handling. In parallel, the emotion-identification component searches the text for keywords associated with the supported emotional categories.

The memory-reel generator then processes the selected recent memories and calculates their average polarity. It applies the predefined decision rules to determine the reel's tone and title and generates a narrative using the available templates. The resulting presentation includes highlights of recent memories.

The integration allows ROOTS to convert stored textual experiences into a structured memory presentation using a single lightweight processing pipeline. The system does not currently use mood-check-in scores as a fully integrated input to the reel-generation algorithm. Similarly, image captioning, speech transcription using an operational speech-recognition model, and LLM-based narrative generation are not part of the implemented pipeline.

The following summarizes the responsibilities of the individual components. Component Input Processing method Output Sentiment analysis Textual memory Keyword scoring and simple negation handling Polarity score in $[-1, 1]$ Emotion identification Textual memory Predefined emotion keyword dictionaries Recognized emotional categories Memory aggregation Recent memory entries Average of available polarity scores Average sentiment polarity Reel selection Average polarity and applicable emotion information Predefined decision rules Reel tone and title Narrative generation Selected reel information and memory content Template-based generation Memory-reel narrative and recent-memory highlights

Table 3. Inputs, processing methods, and outputs of the ROOTS AI pipeline.

4.6 Methodological Limitations

The current AI methodology is designed to demonstrate lightweight memory analysis and automated memory presentation within an integrated application. Several limitations must be considered when interpreting its outputs.

First, the sentiment-analysis component depends on predefined keyword dictionaries and fixed scoring contributions. It may not interpret context, sarcasm, ambiguous expressions, or complex negation accurately. Second, emotion identification depends on whether the memory contains words represented in the predefined categories. An unrecognized emotion may therefore remain undetected.

Third, the memory-reel generator uses average polarity and deterministic selection rules. Consequently, its selected tone and title may not fully represent the diversity of experiences contained in the selected memories. Template-based narration also limits the range and contextual richness of the generated text.

Finally, no quantitative AI evaluation has been reported for the current implementation. There is no established benchmark for sentiment accuracy, emotion-classification performance, or the quality of generated memory reels. The six automated API tests described for the prototype assess selected application workflows; they do not establish the accuracy or psychological effectiveness of the AI component.

Future work can address these limitations through evaluation against manually annotated memory text, comparison with trained sentiment and emotion-classification models, and user studies assessing the usefulness and appropriateness of generated memory presentations. More advanced language processing may also be introduced while preserving the requirement that generated narratives remain grounded in the user's recorded experiences.

5 Implementation and Experimental Setup

5.1 Technology Stack

ROOTS is implemented using React Native, Expo, and TypeScript for the frontend, with Python and FastAPI providing backend services. SQLite is used for persistent storage, with SQLAlchemy managing database interactions. Pytest is used for automated backend testing.

5.2 Implementation

The backend implements five modules: memory management, mood tracking, connection games, memory capsules, and AI processing. REST API endpoints support communication between the frontend and backend. The AI engine performs dictionary-based sentiment analysis, keyword-based emotion identification, and rule-based memory-reel generation.

The prototype supports core memory operations, mood check-ins, Memory Match session persistence, invite-code-based capsules, and AI endpoints. Timeline Challenge results are not persisted through the backend. Voice transcription, image captioning, and LLM-based narration remain unimplemented.

5.3 Testing Procedure

Functional validation uses six automated pytest test categories covering the health endpoint, memory CRUD operations, mood tracking, game sessions, capsules, and AI

endpoints. Manual endpoint checks and application demonstrations supplement these tests.

These tests assess selected functional workflows rather than AI accuracy, application-wide reliability, or user satisfaction.

6 Results and Discussion

6.1 Functional Validation

The prototype demonstrates the integration of memory management, mood tracking, connection games, shared memory capsules, and rule-based AI processing within a single application. The implemented API functionality supports the principal workflows described in the system design.

The six automated test categories provide a basis for checking selected backend operations. The test outcomes should be reported using the actual execution results; individual pass counts or performance measurements should not be inferred without recorded evidence.

6.2 AI Processing

The AI component generates sentiment scores using predefined keyword rules, identifies supported emotional categories, and selects memory-reel titles and tones through deterministic conditions. Template-based narration provides a lightweight method of presenting recent memories without external generative-model inference.

However, no quantitative evaluation of sentiment accuracy, emotion-classification performance, or narrative quality has been conducted. The results therefore demonstrate implemented functionality rather than validated AI performance.

6.3 Limitations and Discussion

The prototype's main contribution is the integration of several memory-related features within a modular mobile application. Its lightweight architecture supports further development, but the current AI methods have limited contextual understanding.

Additional limitations include incomplete voice and image-processing capabilities, partial game-result persistence, and the absence of a complete authentication and authorization framework. No user study has been conducted to determine whether ROOTS improves social connection, reduces homesickness, or affects student well-being.

7 Conclusion and Future Work

7.1 Conclusion

This paper presented ROOTS, an AI-powered memory companion designed for the college transition. The application integrates personal memory management, mood tracking, connection games, memory capsules, and a rule-based AI processing pipeline.

Its modular architecture combines a React Native frontend, FastAPI backend, and SQLite persistence.

Functional validation covers selected API workflows, while the AI component demonstrates keyword-based sentiment analysis, emotion identification, and template-based memory-reel generation. The current work establishes an implemented prototype rather than evidence of improved psychological or social outcomes.

7.2 Future Work

Future development will focus on integrating speech-to-text transcription, image captioning, and language-model-based narration grounded in recorded memories. Further improvements include stronger integration between mood history and memory analysis, more robust sentiment and emotion models, persistent Timeline Challenge results, and authentication with fine-grained access control.

Future evaluation should include quantitative testing of the AI components and a pilot study with college students to assess usability, perceived usefulness, and the suitability of generated memory presentations.

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